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Associations of HbA1c and fasting plasma glucose with incident diabetes: Implications for pre-diabetes thresholds in a Japanese population

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ABSTRACT

Aims: This study assessed pre-diabetes (pre-DM) cutoffs for HbA1c and fasting plasma glucose (FPG) that were associated with an increased risk of incident DM.

Methods: We evaluated 2267 non-diabetic Japanese health-check examinees (HbA1c: <6.5% [<48 mmol/mol] and FPG: <7.0 mmol/L) who were 30–79 years old and were followed-up for 5 years. Incident DM was defined as HbA1c of ≥6.5% (≥48 mmol/mol), FPG of ≥7.0 mmol/L, or physician-diagnosed DM.

Results: During 11047 person-years, we identified 99 incident DM cases (4.3%). The incidence of DM increased with increasing baseline HbA1c or FPG levels, and the change points (95% confidence intervals) were 5.7% (5.6–5.7%; 39 mmol/mol [38–39 mmol/mol]) for HbA1c and 5.5 mmol/L (5.5–5.6 mmol/L) for FPG. The adjusted hazard ratios (HRs) for incident DM per one standard deviation-increase in HbA1c and FPG were 5.5 (4.4–6.8) and 4.0 (3.2–4.8), respectively. The adjusted HRs for incident DM were significantly higher at HbA1c of 5.7–6.4% (39–46 mmol/mol) or FPG of 5.5–6.9 mmol/L, compared to HbA1c of <5.7% (<39 mmol/mol) or FPG of <5.5 mmol/L.

Conclusion: The lower cut-offs for pre-DM may be 5.7% (39 mmol/mol) for HbA1c and 5.5 mmol/L for FPG in this Japanese population.

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1. Introduction

Pre-diabetes is a term that describes an intermediate stage between normal glucose status and diabetes, and cut-off values for glucose levels that are related to an increased risk of incident diabetes have been proposed [1–3]. Therefore, the pre-diabetes state can be used to predict a high risk of incident diabetes and to identify people for whom primary prevention of type 2 diabetes should be considered. Before the introduction of glycated hemoglobin (HbA1c) as a diagnostic criterion [4–7], HbA1c was considered as a significant

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predictor of diabetes in people without diabetes [8–14], especially in people with impaired fasting glucose (IFG; fasting plasma glucose [FPG] levels of 6.1–6.9 mmol/L) [9,13], or people with impaired glucose tolerance (IGT; plasma glucose levels of 7.8–11.0 mmol/L at 2 h after a 75-g oral glucose tolerance test [OGTT]) [14].

In 2009, the International Expert Committee [4] proposed using HbA1c levels of 6.0–6.4% (42–46 mmol/mol) as a definition for impaired glycemic status. The American Diabetes Association (ADA) also recognized that HbA1c levels of 6.0–6.4% (42–46 mmol/mol) were associated with a high risk of incident diabetes, although the ADA adopted a much lower cut-off (5.7%, 39 mmol/mol) [5]. The reasons for the lower cut-off were: (1) a higher probability of identifying IFG or IGT among people with HbA1c levels of <6.0% (42 mmol/mol); (2) HbA1c levels of 5.6% (38 mmol/mol) and 5.4% (36 mmol/mol) corresponding to FPG levels of 6.1 mmol/L and 5.5 mmol/L, respectively, among people without diabetes; and (3) mean HbA1c levels of 5.9% (42 mmol/mol) being observed among people who did not develop diabetes in the Diabetes Prevention Program. Based on the available Japanese data, the Japan Diabetes Society (JDS) proposed using HbA1c levels of 5.6% (38 mmol/mol) and FPG levels of 5.5 mmol/L as the lower limits for identifying a high risk of developing diabetes in the future (i.e., pre-diabetes) [6,15]. Therefore, the present study aimed to examine lower pre-diabetes cut-off values for HbA1c and FPG to determine where the risk of incident diabetes significantly increased in a Japanese population.

2. Methods

2.1. Study population

A total of 3,804 men and 2,208 women (20–86 years old) underwent baseline health check-ups at the Saitama-ken Saiseikai Kurihashi Hospital between February 2006 and January 2007. These people provided informed consent and were followed-up in the Kurihashi Lifestyle Cohort Study, which evaluated the participants' risk factors, diseases, and mortality [16]. The present study's prospective design was approved by our institutional review boards. For the present study, we excluded people who were <30 years old or ≥80 years old (n = 543), people with diagnosed or undiagnosed diabetes based on FPG levels of ≥7.0 mmol/L or HbA1c levels of ≥6.5% (≥48 mmol/mol) (n = 530), people with chronic diseases (e.g., hepatitis, kidney disease, endocrine disease, collagen disease, or cancer; n = 585), and people with missing covariate data (n = 1,584). The remaining 1729 men and 1041 women were 30–79 years old and did not have diabetes based on the FPG and HbA1c criteria (FPG levels of <7.0 mmol/L and HbA1c levels of <6.5% [<48 mmol/mol]). These individuals were considered eligible for the present study, and their data were extracted for analysis. The occurrence of diabetes during the follow-up was defined based on a physician diagnosis of diabetes or test data that exceeded the cut-offs for FPG (≥7.0 mmol/L) or HbA1c (≥6.5%, ≥48 mmol/mol), which were based on the ADA, JDS, and World Health Organization (WHO) diagnostic cut-off values [5–7].

Individuals were initially stratified into quartiles based on HbA1c levels of <6.0% (<42 mmol/mol) or FPG levels of <6.1 mmol/L. This was because HbA1c levels of 6.0% (42 mmol/mol) are the lower limit for a high-risk group according to the International Expert Committee guidelines [4], and because FPG levels of 6.1 mmol/L are defined as the lower limit for IFG according to the WHO [17] and JDS guidelines [6]. However, we found that the numbers of incident diabetes cases were low in the lowest three categories, and we combined these three categories for HbA1c and FPG. Thus, individuals in the present study were categorized based on HbA1c levels of <5.7% (<39 mmol/mol), 5.7–5.9% (39–41 mmol/mol), and 6.0–6.4% (42–46 mmol/mol). Similarly, the individuals were categorized based on FPG levels of <5.5 mmol/L, 5.5–6.0 mmol/L, and 6.1–6.9 mmol/L.

2.2. Outcome measures

The participants completed self-administered questionnaires (use of diabetes medication, diabetes diagnosis, family history of disease, smoking status, and alcohol consumption habits) and underwent a physical examination, laboratory testing, and imaging tests. Smoking status was categorized as never/past smoker or current smoker, and alcohol consumption habits were categorized as never/irregular drinker or regular drinker. The individuals' height, weight, and blood pressures were measured, and body mass index (BMI) was calculated as weight divided by height squared (kg/m^2). Blood pressure was measured in the seated position after 5 min of rest using an automatic sphygmomanometer (Medical Electronics Sphygmomanometer TM-2655P; A&D Co. Ltd.).

Blood samples were drawn in the morning after a 10-h fast, and all biochemical measurements were immediately performed in the hospital laboratory using a VITROS 5.1 FS system (Ortho Clinical Diagnostics, Tokyo). FPG was measured using VITROS GLU slides (Ortho Clinical Diagnostics, Tokyo), based on the glucose oxidase method. Serum total cholesterol and high-density lipoprotein (HDL) cholesterol were measured using VITROS CHOL slides and VITROS dHDL slides, respectively (Ortho Clinical Diagnostics, Tokyo), based on the cholesterol oxidase method and the direct method, respectively. Serum triglycerides were measured using VITROS TRIG slides (Ortho Clinical Diagnostics, Tokyo), based on the lipase/glycerol kinase/dye binding method. HbA1c was measured using the high-performance liquid chromatography method (Adams A1c HA-8160; Arkray, Kyoto). The HbA1c values were reported after conversion from JDS values to the National Glycohemoglobin Standardization Program values (%) and International Federation of Clinical Chemistry and Laboratory Medicine values (mmol/mol) [6]. The mean coefficients of variation were 0.96% and 1.16% at HbA1c levels of 5.8% (40 mmol/mol) and 7.0% (53 mmol/mol), respectively. The mean coefficients of variation were 0.88% and 1.14% at FPG levels of 5.0 mmol/L and 10.0 mmol/L, respectively.

2.3. Statistical analysis

Cox proportional hazards models were used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs) for incident diabetes according to baseline HbA1c or FPG

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Table 1 – The baseline characteristics and follow period of people who did and did not develop diabetes during the follow-up.

	All	People who developed DM	People who did not develop DM	P-value
Number (%, incidence)	2267	99 (4.3)	2168	
Men (%)	63.8	72.7	63.4	0.069
Age (years)	51 ± 9	53 ± 7	51 ± 9	<0.001
Body mass index (kg/m^2)	23.2 ± 2.9	24.6 ± 3.5	23.1 ± 2.8	<0.001
Systolic blood pressure (mmHg)	126 ± 17	134 ± 18	125 ± 17	<0.001
Diastolic blood pressure (mmHg)	76 ± 12	82 ± 12	76 ± 12	<0.001
HbA1c (%)	5.5 ± 0.3	6.0 ± 0.3	5.5 ± 0.3	<0.001
HbA1c (mmol/mol)	37 ± 3	42 ± 3	37 ± 3	<0.001
HbA1c 5.7–6.4% (39–46 mmol/mol) (%)	25.7	87.9	20.9	<0.001
HbA1c 6.0–6.4% (42–46 mmol/mol) (%)	6.9	58.6	4.5	<0.001
FPG (mmol/L)	5.19 ± 0.52	6.01 ± 0.57	5.15 ± 0.48	<0.001
FPG 5.5–6.9 mmol/L (%)	20.6	77.8	18.0	<0.001
FPG 6.1–6.9 mmol/L (%)	5.9	50.5	3.9	<0.001
HDL cholesterol (mmol/L)	1.49 ± 0.39	1.31 ± 0.34	1.50 ± 0.40	<0.001
Family history of diabetes (%)	11.4	19.2	11.1	0.022
Current smoker (%)	27.1	35.4	26.7	0.027
Regular alcohol drinker (%)	32.3	43.4	32.2	0.047
Physically active during leisure time (%)	18.8	16.2	19.0	0.599
Follow period (person-years)	11047	295	10752	

Data are mean ± standard deviation or number and percentage. DM, diabetes mellitus; HbA1c, glycated hemoglobin A1c; FPG, fasting plasma glucose; HDL, high-density lipoprotein.

levels. Before fitting the multivariable model, univariate analyses were used to evaluate each independent variable. The independent variables were FPG (three categories or continuous), HbA1c (three categories or continuous), age (1-year increments), sex, family history of diabetes (yes/no), systolic blood pressure (SBP; 10-mmHg increments), HDL cholesterol (1-mmol/L increments), BMI (1-kg/m^2 increments), alcohol consumption (regular drinker) and smoking status (current smoker). In the models where FPG or HbA1c were treated as a continuous variable, HRs for incident diabetes were associated with a one standard deviation-increase in the levels of FPG or HbA1c. In the models where FPG or HbA1c were treated as a categorical variable, the values of HbA1c or FPG were coded as 1–3 (lowest: 1, highest: 3); category 1 was defined as the reference group. The proportional hazards assumption was tested for each independent variable by plotting the Schoenfeld residuals versus log(time).

The change-points for HbA1c and FPG (where the incidence of diabetes increased significantly) were estimated in separate models using the maximum likelihood method [18]. The 95% CIs for the change points were estimated using bootstrapping (1000 times), and the 2.5% and 97.5% quantiles of the estimated change points were defined as the lower and upper limits of the 95% CIs.

Receiver operating characteristic (ROC) curves for incident diabetes were plotted based on the baseline levels of HbA1c or FPG, and were used to calculate the sensitivity, specificity, and positive predictive value (PPV) for incident diabetes at the different cut-offs. Analysis of variance and the chi-square test were used to compare the characteristics of the various subgroups. All analyses were performed using R software (version 3.1.2; R Foundation for Statistical Computing), and differences were considered statistically significant at a two-tailed P-value of <0.05.

3. Results

3.1. *Baseline characteristics*

Among the 1729 men and 1041 women, 1446 men and 821 women completed the 5-year follow-up and their data were analyzed (Table 1). The follow-up rate was 81.8%, and the incidence of diabetes was 4.3% (99/2267) during 11047 person-years (Table 1). At baseline, >60% of the participants were men, and the mean age was 51 years. The percentages of people with HbA1c levels of 5.7–6.4% (39–46 mmol/mol) and 6.0–6.4% (42–46 mmol/mol) were 25.7% and 6.9%, respectively. The corresponding values for FPG levels of 5.5–6.9 mmol/L and 6.1–6.9 mmol/L were 20.6% and 5.9%, respectively. People who developed diabetes had significantly higher glucose indices and higher percentages for a family history of diabetes, current smoking, and regular alcohol drinking at baseline, compared to people who did not develop diabetes (Table 1).

3.2. *Effects of baseline HbA1c or FPG on the 5-year incidence of diabetes*

The incidence of diabetes increased with increasing baseline levels of HbA1c or FPG (Fig. 1a, b). The change-points were 5.7% (95% CI: 5.6–5.7%; 39 mmol/mol [38–39 mmol/mol]) for HbA1c and 5.5 mmol/L (5.5–5.6 mmol/L) for FPG. The multivariable adjusted HRs for incident diabetes that were associated with a one-standard deviation increase in HbA1c and FPG were 5.45 (95% CI: 4.36–6.83) and 3.95 (95% CI: 3.23–4.83), respectively. In the model where glucose indices were treated as categorical variables, the adjusted HRs for incident diabetes were significantly higher in the groups with HbA1c levels of 5.7–6.4% (39–46 mmol/mol) or FPG levels of 5.5–6.9 mmol/L, compared

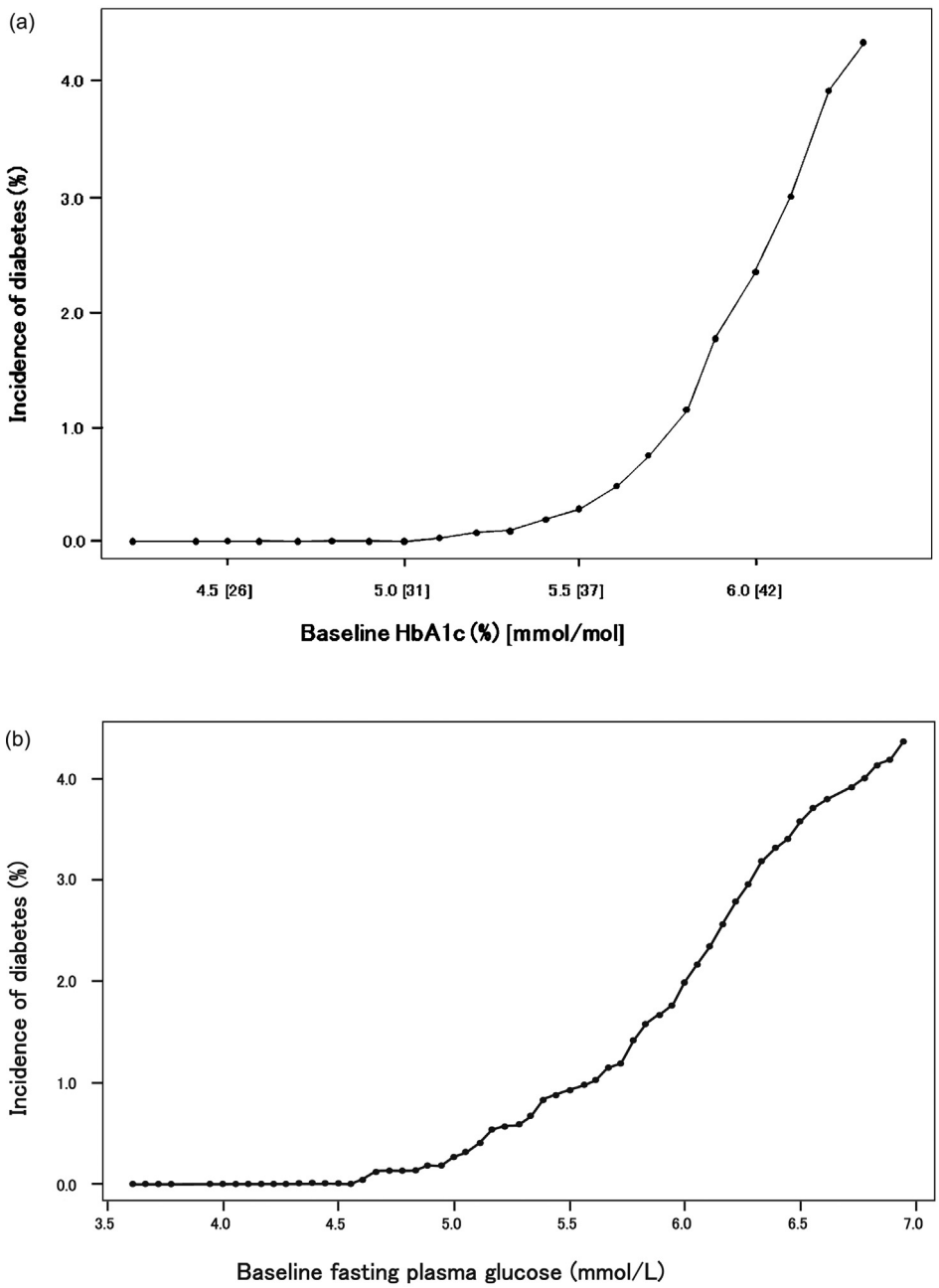

Fig. 1 – Incident diabetes based on (a) glycated hemoglobin A1c (HbA1c) and (b) fasting plasma glucose (FPG) levels at baseline. The change points for predicting diabetes were HbA1c of 5.7% (95% confidence interval: 5.6–5.7%; 39 mmol/mol [38–39 mmol/mol]) and FPG of 5.5 mmol/L (95% confidence interval: 5.5–5.6 mmol/L).

to the groups with HbA1c levels of <5.7% (<39 mmol/mol) or FPG levels of <5.5 mmol/L (Table 2).

3.3. *Predicting incident diabetes using HbA1c or FPG*

The area under the ROC curves (AUCs) were 0.904 (95% CI: 0.870–0.938) for HbA1c and 0.867 (95% CI: 0.825–0.908) for FPG (Supplementary Figure). Using an HbA1c value of 5.7% (39 mmol/mol) gave a sensitivity of 0.879, a specificity of 0.771, and a PPV of 0.149. The corresponding values for HbA1c of 6.0% (42 mmol/mol) were 0.586, 0.955, and 0.372, respectively. Using an FPG value of 5.5 mmol/L gave a sensitivity of 0.788, a specificity of 0.794, and a PPV of 0.149. The corresponding values for FPG of 6.1 mmol/L were 0.505, 0.961, and 0.373.

3.4. *Comparing baseline characteristics according to glycemic status at diabetes onset*

More than 70% (71/99) of incident diabetes cases were identified using the HbA1c cut-off, and more than half (57/99) of incident diabetes cases were identified using the FPG cut-off. The group of incident diabetes cases that were identified using

Table 2 – Hazard ratios and their 95% confidence intervals for three categories of glucose indices that were associated with the development of diabetes.

		Hazard ratio (95% confidence intervals)			Trend P-value
HbA1c categories (%)		<5.7	5.7–5.9	6.0–6.4	
(mmol/mol)		<39	39–41	42–46	
	%, incidence (cases/subjects)	0.7 (12/1684)	6.8 (29/427)	37.2 (58/156)	
	Model 1	1.00	9.72 (4.96–19.05)	63.16 (33.94–117.52)	<0.001
	Model 2	1.00	8.77 (4.47–17.21)	105.47 (29.30–101.86)	<0.001
FPG categories (mmol/L)		<5.5	5.5–6.0	6.1–6.9	
	%, incidence (cases/subjects)	1.2 (21/1743)	7.2 (28/390)	37.3 (50/134)	
	Model 1	1.00	5.94 (3.38–10.47)	37.85 (22.73–63.05)	<0.001
	Model 2	1.00	5.56 (3.09–9.99)	34.89 (19.65–61.95)	<0.001

HbA1c, glycated hemoglobin A1c; FPG, fasting plasma glucose.
Model 1, univariable analysis.
Model 2, multivariable analysis adjusted for age, body mass index, systolic blood pressure, high density lipoprotein cholesterol, smoking habits, alcohol habits, and family history of diabetes.

Table 3 – Comparison of baseline characteristics according to glycemic status at the diagnosis of diabetes.

	Glycemic status at the diagnosis of diabetes		
	FPG DM only (reference group)	HbA1c DM only	FPG and HbA1c DM
Baseline characteristics			
Number	28	42	29
Age (years)	51 ± 7	54 ± 7	53 ± 7
Men (%)	64.3	71.4	82.8
Body mass index (kg/m^2)	24.6 ± 2.6	24.5 ± 3.6	24.9 ± 4.0
Systolic blood pressure (mmHg)	142 ± 19	132 ± 14	132 ± 20
Diastolic blood pressure (mmHg)	86 ± 11	79 ± 10	80 ± 15
HbA1c (%)	5.8 ± 0.3	6.1 ± 0.3*	6.2 ± 0.2*
HbA1c (mmol/mol)	40 ± 3	43 ± 3*	44 ± 2*
FPG (mmol/L)	6.27 ± 0.40	5.70 ± 0.58*	6.21 ± 0.50
HDL cholesterol (mmol/L)	1.42 ± 0.29	1.30 ± 0.38	1.21 ± 0.31
Family history of diabetes (%)	25.0	23.8	6.9
Current smoker (%)	21.4	38.1	44.8
Regular alcohol drinker (%)	46.9	42.9	41.4
Physically active during leisure time (%)	14.3	19.0	13.8
Number of people according to glucose status at baseline			
FPG of <5.5 mmol/L and HbA1c of <6.0% (<42 mmol/mol)	8	11	2
FPG of 5.5–6.9 mmol/L and HbA1c of <6.0% (<42 mmol/mol)	12	5	3
FPG of <5.5 mmol/L and HbA1c of 6.0–6.4% (42–46 mmol/mol)	1	20	7
FPG of 5.5–6.9 mmol/L and HbA1c of 6.0–6.4% (42–46 mmol/mol)	7	6	17

DM, diabetes mellitus; HbA1c, glycated hemoglobin A1c; FPG, fasting plasma glucose; HDL, high-density lipoprotein.
Data are mean ± standard deviation or number and percentage.
* $P < 0.05$ versus people who developed diabetes using FPG alone.
FPG DM only: Incident DM was diagnosed based on FPG of ≥7.0 mmol/L and HbA1c of <6.5% (48 mmol/mol) during the follow-up.
HbA1c DM only: Incident DM was diagnosed based on FPG of <7.0 mmol/L and HbA1c of ≥6.5% (48 mmol/mol) during the follow-up.
FPG and HbA1c DM: Incident DM was diagnosed based on FPG of ≥7.0 mmol/L and HbA1c of ≥6.5% (48 mmol/mol) during the follow-up.

HbA1c had significantly higher mean baseline HbA1c levels, compared to the group that was identified using only FPG. Similarly, the group of incident diabetes cases that were identified using FPG had significantly elevated baseline FPG levels, compared to the group that was identified using only HbA1c (Table 3). No other differences were observed in the baseline characteristics according to the glycemic status at diabetes onset.

4. Discussion

Pre-diabetes was initially defined by the National Diabetes Data Group in 1979 [1], and was adopted by the WHO in 1980 [2] to replace the terms borderline, chemical, and asymptomatic diabetes mellitus. Although people with pre-diabetes rarely exhibit retinopathy and nephropathy [19], they have a higher risk of developing diabetes, compared to individuals with normal glucose levels. Thus, the recommended pre-diabetes cut-offs for FPG and HbA1c have been chosen based on longitudinal data analysis [5–7]. In the present study, we used the same methods as in previous reports, and revealed that the adjusted HRs for incident diabetes increased with increases in the levels of HbA1c or FPG. Furthermore, the HRs were significantly higher in groups with HbA1c levels of ≥5.7% (≥39 mmol/mol) or FPG levels of ≥5.5 mmol/L, compared to the lowest HbA1c (<5.7% [<39 mmol/mol]) or FPG (<5.5 mmol/L)

categories. Moreover, these cut-off values were in accordance with the values from the change-point analysis, which has not been performed in previous Japanese studies. Therefore, our findings support the validity of the current HbA1c and FPG thresholds for high-risks groups from the international and Japanese guidelines [4–6].

A 5-year prospective and collaborative study in Denmark, Australia, and France revealed variations in the HbA1c thresholds for incident diabetes as defined using treatment, HbA1c levels, or FPG levels, based on change-point analysis: 5.1% (95% CI: 4.9–5.6%; 32 mmol/mol [30–38 mmol/mol]), 5.4% (95% CI: 5.1–5.6%; 34 mmol/mol [32–38 mmol/mol]), and 5.3% (95% CI: 5.1–5.7%; 34 mmol/mol [32–39 mmol/mol]) [21]. Ethnicity, age, genetic background, erythrocyte life span, and environmental factors may influence inter-individual and inter-group variability in hemoglobin glycosylation [22–26], and may explain the different cut-offs for HbA1c in the different studies. Furthermore, these change-points were calculated using adjusted regression models. Thus, it is impossible to simply compare these values with our values. Similar variations in the FPG change-points were also reported in the same collaborative study [21]: 5.1 mmol/L (95% CI: . . .–6.1 mmol/L) in Denmark, 5.5 mmol/L (95% CI: 5.2–5.5 mmol/L) in Australia, and no-change point in France. In this context, change points are influenced by the observation period and may become lower with longer follow-ups [27]. However, the available data (including ours) were obtained using the same short follow-up. Furthermore, it is difficult to anticipate what 5-year periods in each participant's latency period were selected for the analysis. Therefore, longer follow-ups and repeated measures data analysis may be necessary to confirm the validity of the lower thresholds.

The possible lowest levels for defining pre-diabetes using HbA1c and FPG have been explored in prospective studies using Cox models [10–12,15,28,31–35] and ROC curve analysis [28–35]. In the Japanese studies, the thresholds for HbA1c exhibited noticeable variations (5.3% [34 mmol/mol] to 5.8% [40 mmol/mol]) [10–12,15,28], although the range for FPG was relatively small (5.5–5.7 mmol/L) [12,15,32–35]. Our results are within these expected ranges, and the overall abilities of HbA1c or FPG levels to predict incident diabetes were similarly high in the current study, compared to the abilities from previous reports [12,15,28–35].

Since the introduction of pre-diabetes, based on FPG levels, as IFG by the ADA [20,36], there has been debate regarding whether the lowest limit of FPG for IFG should be lowered from 6.1 mmol/L to 5.5 mmol/L [31–37]. In the present study, lowering the IFG threshold from 6.1 mmol/L to 5.5 mmol/L increased the prevalence by 3.7-fold, and similar results have been reported in other studies [14,15,31–37]. Lowering the HbA1c threshold for pre-diabetes from 6.0% (42 mmol/mol) to 5.7% (39 mmol/mol) also increased the prevalence by 3.5-fold. In this context, people with pre-diabetes who would be identified using the lowered thresholds have a lower risk of incident diabetes, compared to people with pre-diabetes who are identified using a higher threshold. However, the risk of incident diabetes started to significantly increase at the lower thresholds, which were more sensitive than the higher thresholds for identifying individuals at an elevated risk of incident diabetes.

The current study revealed that FPG and HbA1c levels provided poor agreement as screening tests for pre-diabetes, which confirms the findings of previous reports [29,30,38]. Interestingly, people who developed diabetes using only FPG levels exhibited higher baseline FPG levels and lower baseline HbA1c levels, compared to people who developed diabetes using only HbA1c levels. The same trend was also observed when we compared diabetes cases that were diagnosed using only HbA1c levels or only FPG levels. However, there were no clear differences in the characteristics of the incident cases of diabetes that were diagnosed using only FPG or HbA1c levels. Thus, the combined use of FPG and HbA1c levels may be reasonable to avoid missing people with a high risk of incident diabetes, and it is even efficient for identifying people with the highest risk among people with pre-diabetes. The JDS recommends OGTT for people with HbA1c levels of 5.6–6.0% (38–42 mmol/mol) or FPG levels of 5.5–6.0 mmol/L, in order to avoid missing people with a high risk of developing diabetes, and especially people with IGT who are targeted for lifestyle interventions [6]. Unfortunately, we did not have access to OGTT data, and could not consider people with IGT in our analyses.

The present study has several limitations. First, this study analyzed data that were obtained from Japanese individuals at their health check-ups, and the findings may not generalize to other populations or other groups of people. Nevertheless, few studies have identified pre-diabetes cut-offs for HbA1c using longitudinal and annually repeated data measurements. Second, this study did not include OGTT data, and there is no information regarding diabetes that was diagnosed based on 2-h plasma glucose, which could have resulted in misclassifications of individuals. Therefore, longitudinal data that include OGTT findings are needed. Third, we excluded 36% of the potentially eligible people, as they were missing covariate data, despite the fact that they did not have diabetes or life-threating conditions. Furthermore, the people who were included in the analysis had slightly healthier profiles, compared to the excluded individuals. Thus, the possibility of selection bias should be considered, which could have lowered the apparent incidence of diabetes and may make it difficult to identify solid cut-off values in this population. Fourth, we did not consider the influence of hemoglobinopathies, although these conditions are very rare in Japan [39] and likely had minimal effects on our findings.

In conclusion, the findings from this 5-year follow-up study of Japanese individuals indicate that the implied lower HbA1c and FPG cut-offs for defining pre-diabetes in this population were 5.7% (39 mmol/mol) and 5.5 mmol/L, respectively.

Conflicts of interest

All authors have no conflicts of interest to disclose.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.pcd.2016.07.006.

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